Chapter1: Electricity and magnetism

Electric charge:

The property of a proton or an electron ,which gives rise to electrostatics force between them is called electric charge.

There are two types of electric charge:

- 1.Positive charge(Carried by proton)
- 2.Negative charge (Carried by electron)

Like charge(Repel)

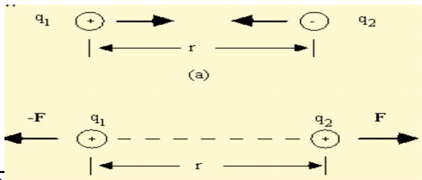
Unlike charge(Attractive)

Neutral (Net charge absence)

Coulomb's Law for electrostatics:

The force between two point charge at rest is directly proportional to the product of the magnitude of the charge q_1 and q_2 is inversely proportional to the square of the distance between them,

F$\propto q_1\, q_2$



F$\propto \frac{1}{r^2}$

F$\propto \frac{q_1\, q_2}{r^2}$

F$\propto K\, \frac{q_1\, q_2}{r^2}$

In the case of vacuum and using S.I. units, the value

$$k = 9.0 \times 10^9 \ \mathrm{Nm^2\, C^{-2}} = \frac{1}{4\pi\varepsilon_0}$$

Where ε_0 is the permittivity constant for vacuum and the value is

$$\varepsilon_0 = 8.854\times10^{-12}\ C^2 N^{-1} m^{-2}$$

Therefore, in vacuum Coulomb's law can be represented by

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{d^2}$$

For any other medium it can be represented by

$$F = \frac{1}{4\pi\varepsilon}\frac{q_1 q_2}{d^2}$$

Point Charge:

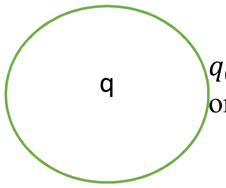
Any charge which covers a space with dimensions much less then its distance away from an observation point can be consider a point charge.

The electric field:

When an electric charge is placed at same point in space,this establish every where a state of electric stress which is called an electric field.

Intensity of electric field: $\overline{E}$

$$F= \frac{1}{4\pi\varepsilon_0}\frac{q_1\,q_2}{r^2} \ldots\ldots\ldots\ldots\ldots\ldots(1)$$



q_0(attractive force or repel force)

Equation 1 dividing by q_0,

$$\overline{E}=\frac{F}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$$

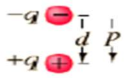
Electric potential:

The electric potential at a point in electric field is defined as the work done in bring a unite positive charge from infinity to that point in a question, if W be the work done in bringing a positive charge from infinity to the point then electric potential,

$$V=\frac{W}{q_0}$$

Electric dipole:

When two equal and opposite charges placed at a small distance apart, then the arrangement is called an electric dipole.



$$\vec{p} = q\vec{d}$$

Potential and Field due to an electric Dipole,

$$V = \frac{1}{4\pi\varepsilon_o}\left[\frac{q}{r_1} - \frac{q}{r_2}\right] \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots.(1)$$

If $r \gg l$, then r_1 and r_2 may be written as

$$r_1 = r - l\cos\theta$$
$$r_2 = r + l\cos\theta$$

Substituting these values in eqn. (1); we get

$$V = \frac{1}{4\pi\varepsilon_o}\left[\frac{q}{r - l\cos\theta} - \frac{q}{r + l\cos\theta}\right]$$

$$= \frac{1}{4\pi\varepsilon_o}\left[\frac{q.2l\,\cos\theta}{r^2 - l^2\cos\theta}\right]$$

$$= \frac{1}{4\pi\varepsilon_o}\frac{2ql\,\cos\theta}{r^2}$$

$$= \frac{1}{4\pi\varepsilon_o}\frac{p\cos\theta}{r^2} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(2)$$

The electric field to P due to the dipole may now be evaluated .It is convenient to use polar co-ordinates (r,θ) in place of Cartesian co-ordinates (x,y) for evaluating the components E_r and E_θ of the electric field are given by

$$E_r = -\frac{\partial V}{\partial r} = -\frac{\partial}{\partial r}\left[\frac{1}{4\pi\varepsilon_o}\frac{p\cos\theta}{r^2}\right] = \frac{1}{4\pi\varepsilon_o}.\frac{2p\cos\theta}{r^3} \ldots\ldots\ldots\ldots (3)$$

$$E_\theta = -\frac{1}{r}\frac{\partial V}{\partial\theta} = -\frac{1}{r}\frac{\partial}{\partial\theta}\left[\frac{1}{4\pi\varepsilon_o}\frac{p\cos\theta}{r^2}\right] = \left[\frac{1}{4\pi\varepsilon_o}.\frac{p\sin\theta}{r^3}\right] \ldots\ldots (4)$$

The magnitude of total electric field is given by

$$E = \sqrt{(E_r^2 + E_\theta^2)} = \frac{1}{4\pi\varepsilon_o}\sqrt{\left[\left(\frac{2p\cos\theta}{r^3}\right)^2 + \left(\frac{p\sin\theta}{r^3}\right)^2\right]}$$

$$= \frac{1}{4\pi\varepsilon_o}.\frac{p}{r^3}\sqrt{(1+3\cos^2\theta)} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (5)$$

The angle α which the electric field vector $\vec{E}$ makes with radius vector $\overrightarrow{OP}$ is given by

$$\alpha = tan^{-1}\left(\frac{E_\theta}{E_r}\right) = tan^{-1}\left(\frac{1}{2}\tan\theta\right)$$

It is obvious from equations (2), (4) and (5) that field and potential of a dipole vary as $\frac{1}{r^3}$ and respectively contrary to the laws of a single charge where the electric field and potential vary as $\frac{1}{r^2}$ and $\frac{1}{r}$ respectively.

Electric potential due to point charge:

3-3. Electric Potential due to Point Charge. Let us consider an isolated positive point charge q placed at the origin O as shown in Fig. 3.3. We want to calculate its electric potential at a point P at distance r_A from it. Suppose a test charge q_0 is placed at distance r_B from O.

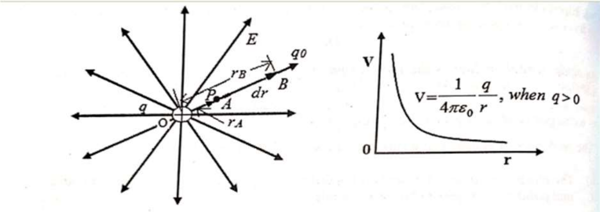


Fig. 3.3 Potential difference between two points due to a point charge q.

Hence the force F acts away from the charge q. The small work done in moving the test charge q_0 from 'B' to 'A' through a small displacement dr against the electrostatic force is

$$dW = \vec{F}.d\vec{r} = Fdr\cos 180^o = -Fdr$$

By Coulomb's law, the electrostatic force acting on charge q_0 is

$$F = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2}$$

The total work done in moving the test charge q_0 from infinity to the point P is

$$W = \int_B^A dW = -\int_B^A \frac{1}{4\pi\varepsilon_0}.\frac{qq_0}{r^2}dr$$

$$= -\frac{qq_0}{4\pi\varepsilon_0}\int_{r_B}^{r_A} r^{-2}dr$$

If we choose the point B to be at infinity, $r_B \to \infty$ and $V_B = 0$, then

$$= -\frac{qq_0}{4\pi\varepsilon_0}\left[-\frac{1}{r}\right]_\infty^r$$

$$= \frac{qq_0}{4\pi\varepsilon_0}\left[\frac{1}{r} - \frac{1}{\infty}\right]$$

$$= \frac{qq_0}{4\pi\varepsilon_0}\frac{1}{r}$$

$$V = V_A - V_B = \frac{W_{BA}}{q_0}$$

$$\therefore V_A - V_B = \frac{qq_0}{4\pi\varepsilon_0}\frac{1}{r}.\frac{1}{q_0}$$

$$\therefore V_A = \frac{1}{4\pi\varepsilon_o}\times\frac{q}{r_A}$$

From the above equation it is clear that for a given charge, the potential depends only upon *r*. So, electric potential has spherical symmetry. If the source charge is positive, *i.e., if $q > 0$, then V is positive, similarly, if q is negative, then potential is negative.*

Flux of the electric field(∅):(Shine)

একটি particular area এর মধ্যে দিয়ে কি পরিমান electric field প্রবাহিত হচ্ছে এই জিনিসটার মানকে Flux (∅) বলে।

The flow through an flux an element of surface is just equal to the component of the velocity perpendicular to the surface times the area of the surface.

Electric flux over a closed surface:

$d\emptyset = \overline{E}\cos\alpha \times ds$

$= \overline{E}\,\hat{n}\,ds$

$= \overline{E}\,\overline{ds}$

For total surface,

$\emptyset = \oint \overline{E}\,\overline{ds}$

Electric flux density:

$\overline{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$

$\overline{\varepsilon_0 E} = \frac{1}{4\pi}\frac{q}{r^2}\hat{r}$

$\boxed{\vec{D} = \frac{1}{4\pi}\frac{q}{r^2}\hat{r}}$

Gauss's law:

The flux of the electric field $\vec{E}$ through any closed surface i.e the integral $\oint \vec{E}.d\vec{s}$ over the surface is equal to $\frac{1}{\varepsilon_0}$ times the total charge enclosed by the surface and if the surface dosen't enclosed the charge the flux of $\oint \vec{E}.d\vec{s}$ equal to zero.

$\oint \vec{E}.d\vec{s} = \frac{q}{\varepsilon_0}$ (When q inside the surface)

$\oint \vec{E}.d\vec{s} = 0$ (when q outside the surface)

Let us calculate total flux at point o,

$\emptyset = \oint \underline{E}.d\underline{s}$

$= \oint \underline{E}.d\underline{s_1} + \oint \underline{E}.\ d\underline{s_2} + \oint \underline{E}.d\underline{s_3}$

$= \int \frac{q ds_1 \cos\theta_1}{4\pi\varepsilon_0\, r^2} - \int \frac{q ds_2 \cos\theta_2}{4\pi\varepsilon_0\, r^2} + \int \frac{q ds_3 \cos\theta_3}{4\pi\varepsilon_0\, r^2}$

We know that,

$= \frac{q}{4\pi\varepsilon_0}\{(dw_1 - dw_2 + dw_3)\}$ [where, $dw = \frac{ds\cos\theta}{r^2}$]

$= \frac{q}{4\pi\varepsilon_0}\{(dw - dw + dw)\}$

$= \frac{q}{4\pi\varepsilon_0} \times dw$ [where $dw = 4\pi$]

$= \frac{q}{4\pi\varepsilon_0} \times 4\pi$

$= \frac{q}{\varepsilon_0}$ [when inside the surface]

Let us calculate total flux at point p,

$\emptyset = \oint \underline{E}.d\underline{s}$

$= \oint \underline{E}.d\underline{s_1} + \oint \underline{E}.\ d\underline{s_2} + \oint \underline{E}.d\underline{s_3}$

$= \int \frac{q ds_1 \cos\theta_1}{4\pi\varepsilon_0\, r^2} - \int \frac{q ds_2 \cos\theta_2}{4\pi\varepsilon_0\, r^2} + \int \frac{q ds_3 \cos\theta_3}{4\pi\varepsilon_0\, r^2} - \int \frac{q ds_4 \cos\theta_4}{4\pi\varepsilon_0\, r^2}$

We know that,

$= \frac{q}{4\pi\varepsilon_0}\{(dw_1 - dw_2 + dw_3 - dw_4)\}$ [where, $dw = \frac{ds\cos\theta}{r^2}$]

$= \frac{q}{4\pi\varepsilon_0}\{(dw - dw + dw - dw)\}$

$= 0$ [when outside the surface].

Gauss's law law from Coulomb's Law:

$F = \frac{1}{4\pi\varepsilon_0}\frac{q\ \ q_0}{r^2}$

$F = q_0 E \ldots\ldots\ldots\ldots(1)$

$\bar{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$

Or, $\oint E \times 4\pi r^2 = \frac{q}{\varepsilon_0}$

$\therefore \oint \underline{E}.d\underline{s} = \frac{q}{\varepsilon_0}$ [proved]

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